Function Generator



Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **Medium**

1 labs

Covers these lab items: Function generator

What it is used for

A function generator produces electrical waveforms such as sine, square, and triangle signals for testing and demonstrating electronic circuits. Correct amplitude, frequency, and offset settings help verify circuit behaviour safely and accurately.

Main parts

- Function generator unit and display
- Waveform selection and frequency controls
- Amplitude and DC offset controls
- Output terminals or BNC connectors
- Power switch and power supply
- Test leads and compatible connecting cables

Before you start

1. Read the instrument manual and identify the voltage, frequency, and output limits.
2. Inspect the case, display, connectors, power cable, and test leads for damage.
3. Confirm that the circuit under test can tolerate the planned signal amplitude and offset.
4. Check the required output termination and cable type for the test setup.
5. Switch on the generator and allow it to complete its startup checks.
6. Ensure the output is disabled while connecting leads and configuring initial settings.

How to use

1. Select the required waveform, such as sine, square, or triangle.
2. Set the target frequency using the frequency controls or numeric entry.
3. Set the output amplitude and DC offset within the circuit's permitted limits.
4. Select the required output load or termination setting if supported by the instrument.
5. Connect the output cable to the circuit input with the generator output disabled.
6. Check the wiring and confirm the circuit ground and signal connections are correct.
7. Enable the output and observe the circuit response using an oscilloscope or suitable measuring instrument.
8. Adjust frequency, amplitude, or offset gradually while monitoring the response.
9. Disable the output before disconnecting the circuit or changing the wiring.
10. Switch off the generator, remove the leads, and store the accessories safely.

⚠ Safety

- Never exceed the input voltage or signal limits of the circuit or connected instrument.
- Disable the output before connecting, disconnecting, or changing test leads.
- Do not connect the generator output directly to mains voltage or an incompatible power source.
- Use suitable cables and avoid shorting the output terminals.
- Keep the instrument and connectors dry and use appropriate electrical safety practices.
- Disconnect power before opening the case or performing internal maintenance.

Care & storage

- Keep the display, case, and connectors clean using approved materials.
- Inspect output cables and connectors for loose contacts or damaged insulation.
- Store leads neatly and avoid bending cables sharply near their connectors.
- Arrange calibration and functional checks according to the laboratory schedule.

Common mistakes to avoid

- Setting the wrong amplitude or offset can damage the circuit under test.
- Ignoring output termination can produce an incorrect signal amplitude.
- Connecting leads while the output is enabled can create unexpected transients or shorts.
- Using damaged cables can cause unstable signals and unreliable measurements.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.